Gürkan Doğukan Karakurt

GRADUATE RESEARCHER · CIVIL ENGINEER · SOFTWARE DEVELOPER

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MSc candidate and Research Assistant in Construction Engineering & Management at METU, supervised by Prof. Dr. Rıfat Sönmez. Thesis applies graph neural networks to BIM models to generate complete construction precedence schedules from a building's geometry and standardized activity classifications. Background in civil engineering (BSc, METU) with a minor in Corporate Finance and a long-running practice in full-stack software development and graphic design.

EDUCATION

2025 – Present

MSc, Construction Engineering & Management

Middle East Technical University (METU), Ankara

Research Assistant. Thesis on BIM-driven scheduling with graph neural networks, supervised by Prof. Dr. Rıfat Sönmez. Transferred from the Geotechnical Engineering programme in Spring 2025.

2017 – 2023

BSc, Civil Engineering

Middle East Technical University (METU) — GPA 3.35 / 4.00

Coursework in structural analysis, geotechnics, hydraulics, construction management, and Building Information Modeling.

2019 – 2023

Minor, Corporate Finance

Middle East Technical University (METU) — GPA 3.39 / 4.00

Minor programme covering financial analysis, valuation, and quantitative methods alongside the civil engineering degree.

MSC THESIS · CONSTRUCTION ENGINEERING & MANAGEMENT

Predicting Construction Precedence Relationships from BIM Geometry with Machine Learning

SUPERVISOR: PROF. DR. RIFAT SÖNMEZ · CE 520 — GRADUATE SEMINAR, SPRING 2026 · PRESENTED 12 MAY 2026 AT METU

Construction megaprojects miss budget 98 % of the time, largely because BIM and the construction schedule still do not talk to each other. This thesis closes that loop: given a BIM model with assigned OmniClass activities, a graph neural network predicts the entire FS, SS, FF, SF precedence network — regenerating schedules from the model on every design change.

PROBLEM

The 4D BIM loop is broken: schedule logic is not derived from the model itself. Planners rebuild precedence by hand from drawings for each project. Recent text-based approaches (e.g. Amer, Jung & Golparvar-Fard, 2023, ILC) reach F1 91 % on activity pairs, but ignore geometry, hit token-window limits at real project scale, and produce loops and broken links.

APPROACH

A single graph neural network (GraphSAGE / GAT) consumes a three-layer IFC representation: native IFC class and geometry, Uniformat II classification, and OmniClass T22 activities. Message passing over BIM topology — adjacency, support, containment — yields probabilities for every candidate FS / SS / FF / SF link in the schedule. Standardized vocabularies replace free-form text and avoid LLM hallucinations.

CONTRIBUTION

First end-to-end whole-schedule precedence predictor that grounds directly in BIM topology and standardized classification codes, instead of free-form activity text. Designed for transfer across firms and projects, with a roadmap to a defense in December 2027.

EXPERIENCE

Oct 2025 – Present

Research Assistant — Construction Engineering & Management

Middle East Technical University (METU), Department of Civil Engineering

Graduate research and teaching support within the Construction Engineering and Management division. Focus areas include BIM, graph neural networks for construction scheduling, and machine learning applied to the built environment.

Dec 2024 – Oct 2025

Software Engineer

Mega Mühendislik, Türkiye

Developed software for engineering workflows, contributing across the stack to ship reliable internal tooling before moving full-time into graduate research at METU.

2023 – 2024

Head of Quantitative Strategies

Vortex · Remote (Georgia)

Designed and developed algorithmic trading systems for financial markets — scalping and swing strategies, performance tuning, and per-investor mandate customisation.

2021 – 2024

Web Development Project Lead

Self-initiated project

Led a self-started web development project focused on financial markets, delivering a web-based trading interface end to end and owning product decisions across the build.

Summer 2022

Civil Engineering Intern

ES Group — Erdemir Port Project, Zonguldak

Worked across site operations, planning, and progress payments on a port construction project. Contributed to steel-pile design and took responsibility in cost control and project scheduling.

2020 – 2024

Freelance Graphic Designer · 2024 Local Elections Lead Designer

Independent

Delivered posters, brochures, social media content, and campaign materials across freelance projects. Served as lead designer for a 2024 Turkish mayoral campaign, owning the full digital content production pipeline.

SELECTED PROJECTS

Diyet Cebimde — AI-assisted diet & nutrition platform — full-stack, solo-built

2025 — Ongoing · Mobile + Web Admin + API

Mobile platform that generates AI-driven weekly diet plans from each user's profile and goals, with photo-based meal recognition, manual tracking, and a memory-equipped chatbot. Hybrid pipeline: Gemini for meal selection, PuLP linear programming for portion optimisation; snapshot architecture keeps historical plans stable when the food catalog changes.

Stack: Django 5.2 · DRF · PostgreSQL · Gemini API · PuLP · React 18 + TS · Expo / React Native · Railway

SKILLS

Research & Engineering: BIM (Revit, IFC), Construction Mgmt (scheduling, CPM), Graph Neural Networks (GraphSAGE / GAT), Machine Learning (PyTorch), AutoCAD (upper-intermediate), SAP2000 (beginner-intermediate), EPANET (advanced), MS Project (intermediate)

Software Development: Python & Django (advanced), HTML / CSS / JS (advanced), React (intermediate), PostgreSQL (intermediate), MATLAB / VBA (advanced)

Design: Photoshop (advanced), Illustrator (advanced), Premiere Pro (advanced), After Effects (intermediate)

Languages: Turkish (native), English (C1)

Personal contact details (email, phone, address) are intentionally omitted from this CV for privacy. Please reach out through LinkedIn or ORCID for academic correspondence.